

ENGELBERG, Switzerland

WMO: 066550

Lat: 46.8217N

Lon: 8.4106E

Elev: 3402

StdP: 12.98

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.2	12.2	2.7	7.2	9.1	7.3	9.1	14.1	20.1	46.1	15.2	41.3	2.5	100	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	79.9	64.1	76.5	63.1	73.4	61.5	65.7	76.4	64.1	74.1	62.6	71.6	7.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.8	93.8	70.5	60.3	88.6	68.7	59.0	84.6	67.0	32.5	76.6	31.2	74.4	30.0	71.8	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	12.1	9.7	4.2	84.9	4.9	2.6	0.6	86.7	-2.3	88.2	-5.0	89.7	-8.6	91.5
			3.2	68.7	4.8	1.5	-0.2	69.8	-3.0	70.6	-5.7	71.5	-9.2	72.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.1	28.8	29.5	37.1	44.9	51.0	58.6	61.1	60.1	53.2	47.2	37.8	30.9
	DBStd	13.75	8.44	8.86	8.13	7.70	7.19	7.03	6.03	6.02	6.28	7.30	8.38	8.28
	HDD50	3049	658	573	403	186	76	10	2	2	39	137	371	592
	HDD65	7341	1123	993	866	602	435	213	151	176	355	553	817	1057
	CDD50	1260	0	0	2	34	108	268	345	314	135	49	5	0
	CDD65	73	0	0	0	0	1	20	29	22	1	0	0	0
	CDH74	589	0	0	0	0	16	189	219	162	3	0	0	0
	CDH80	85	0	0	0	0	1	28	32	24	0	0	0	0
	WSAvg	3.5	2.9	3.2	3.8	4.4	4.2	3.9	3.6	3.4	3.2	3.1	3.1	3.0
	PrecAvg	64.8	3.7	4.1	4.3	5.0	6.9	7.2	7.2	7.4	5.4	4.2	4.9	4.6
(15) Precipitation	PrecMax	81.5	8.8	12.1	8.5	10.0	10.9	11.1	13.7	16.0	9.2	7.5	13.6	11.4
	PrecMin	49.3	0.2	1.3	1.7	1.1	1.7	4.3	3.1	2.2	2.1	0.2	0.2	0.2
	PrecStd	7.3	2.2	2.4	1.9	2.0	2.8	1.7	2.6	2.9	1.8	1.8	2.7	2.5
	PrecStd	7.3	2.2	2.4	1.9	2.0	2.8	1.7	2.6	2.9	1.8	1.8	2.7	2.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	52.4	51.7	59.7	69.9	76.2	83.8	83.6	83.2	73.6	67.8	60.6	55.5	
	0.4% MCWB	43.0	40.3	45.9	52.8	59.6	64.7	65.8	64.2	61.3	53.2	46.9	42.4	
	2% DB	46.7	47.1	55.9	65.4	71.2	79.3	79.5	78.4	70.1	64.5	57.6	48.3	
	2% MCWB	38.4	38.0	43.6	50.5	56.8	63.5	64.2	63.9	59.4	52.7	45.5	39.8	
	5% DB	42.7	43.9	52.7	61.8	67.7	75.5	75.9	74.6	66.7	60.7	53.1	43.6	
	5% MCWB	37.7	36.4	42.4	48.6	55.0	62.1	63.4	62.8	57.7	51.6	45.0	38.4	
	10% DB	39.5	40.5	49.0	58.0	64.0	71.6	72.6	70.8	63.4	57.3	48.6	40.6	
	10% MCWB	35.6	35.5	40.6	46.6	53.0	60.5	61.6	61.6	56.2	50.7	43.5	36.4	
	0.4% WB	44.1	42.3	47.5	53.9	61.6	67.3	67.8	67.1	63.4	56.9	49.6	43.9	
	0.4% MCDB	50.3	48.2	56.6	66.9	73.5	80.0	79.6	77.4	70.8	62.1	55.4	50.7	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	40.4	39.8	45.1	51.5	58.7	64.9	65.7	65.3	60.8	54.6	47.3	41.7	
	2% MCDB	44.8	44.9	53.4	63.1	68.4	76.0	76.9	75.2	67.6	59.7	53.7	47.2	
	5% WB	37.9	37.7	43.1	49.4	56.0	63.0	64.0	63.5	58.8	53.1	45.6	39.3	
	5% MCDB	41.5	42.3	50.7	59.7	65.1	73.4	74.1	72.5	64.9	58.3	51.7	43.3	
	10% WB	36.0	35.7	41.4	47.6	53.9	61.2	62.4	61.9	57.0	51.5	43.9	37.1	
	10% MCDB	39.4	40.1	48.0	56.5	62.3	70.4	71.1	69.7	62.4	56.5	48.9	40.0	
(35) Mean Daily Temperature Range	MDBR	10.8	12.2	14.9	16.8	17.1	17.3	17.1	16.0	14.5	13.9	11.4	10.5	
	5% DB	14.1	15.7	19.8	22.5	23.6	24.0	22.7	21.9	19.1	18.2	16.4	13.7	
	5% WB	9.1	9.4	11.3	12.1	12.8	12.5	11.6	11.7	11.3	10.1	9.0	8.4	
	MCDBR	12.9	13.9	18.2	21.4	21.2	22.2	21.2	20.1	17.4	15.5	14.8	13.0	
	MCWBR	8.9	9.4	10.9	12.0	12.1	12.0	11.5	11.2	10.7	9.6	9.0	8.7	
	MCWBR	8.9	9.4	10.9	12.0	12.1	12.0	11.5	11.2	10.7	9.6	9.0	8.7	
(40) Clear-Sky Solar Irradiance	taub	0.270	0.300	0.350	0.396	0.416	0.421	0.422	0.409	0.376	0.345	0.302	0.267	
	taud	2.330	2.251	2.179	2.134	2.162	2.203	2.203	2.253	2.327	2.386	2.410	2.358	
	Ebn at Noon	262	274	274	271	270	268	266	265	264	255	247	249	
	Edh at Noon	23	31	39	45	46	44	44	40	34	27	22	20	
	Edh at Noon	23	31	39	45	46	44	44	40	34	27	22	20	
(44) All-Sky Solar Radiation	RadAvg	458	751	1134	1401	1541	1800	1777	1545	1191	777	463	380	
	RadStd	55	93	161	218	153	163	157	144	118	88	65	52	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (119 neighbors)	N/A	N/A	N/A	+1.30	+0.72	+0.56	N/A	N/A	+104	+29

Nomenclature: See separate page